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Study of repeated susceptibility to cement dust exposure and its persistent consequences on cardiovascular markers among male cement handlers at Dalmia Bharat Cement Plant

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Abstract- According to the current study, cement male handlers at Dalmia Bharat Cement Plant had persistent effects of cement dust exposure on certain cardiovascular indicators. This study found that repeated and protracted exposures, depending on sensitivity and duration, have led to detrimental health conditions and deteriorating health, especially in their chosen cardiovascular indicators. The study's conclusions showed that uncontrolled dust exposure caused about considerable increase in determinants of Coronary Artery Disease (CAD) statically caused by macro vascular accelerate atherosclerosis. Results of current study unveiled risk for CAD had been significantly associated with triglyceride total cholesterol, low density lipoprotein, age body mass index, smoking, hypertension ,however risk for CAD was not related with high density lipoprotein. Thus, the present study finds positive association of cement dust exposures with hypertensive and pulmonary cardiac disease among cement handlers (CH) when compared with Non-cement handlers (NCH) as persons control of Dalmia Bharat Cement Plant.

Keywords: Cement dust, Cardiovascular Markers, Coronary Artery Disease, Environmental pollution, Factory workers

INTRODUCTION

An intolerable number of health problems, most frequently associated with atherosclerosis, have always been caused by environmental pollution. According to this perspective, the cement sector is important and is growing quickly both globally and in India. Due to persistently high levels of construction development activity, the emerging market has seen an increase in demand for cement manufacturing.[1-3] The cement industry contributes to pollution in the environment.[4] The many stages of cement manufacture, such as clinker cooling, raw material grinding, rotating kilns, packaging facilities, and storage units, produce dust and other airborne particles.[5-6] Line, silica, alumina, and iron oxides make up the majority of the basic ingredients required to make cement.[7] Silica exposure can cause heart disease and other detrimental effects on many bodily organs; it has been a significant health concern.[8] Exposure to cement dust has been a growing issue and is the cause of coronary artery disease (CAD) and respiratory disorders.[9,10] Additionally, toxic cement dust damages many human organs through inflammation. Cement dust enters our bodies by eating, inhalation, and epidermal exposures, to a lesser degree. Dust susceptibility is linked to major changes in the lipoprotein and plasma lipid profiles as well as an elevated risk of coronary artery disease.[11] A major cause of mortality,

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coronary artery disease is brought on by premature atherosclerosis. Exposure to cement dust among cement handlers or manufacturing workers be connected several intentional hazard factors, that includes hypertriglyceridemia, hypercholesterolemia, and hypertension.

MATERIAL & METHODS

The study involved a total of 120 participants from the Dalmia Bharat Cement Plant located in Kalyanpur, Banjari, Bihar. Among participants, 60 were workers directly exposed to cement dust, referred to as cement handlers (CH), due to their job responsibilities over the past eight years. The remaining 60 participants, who had no exposure to cement dust, served as controls and were classified as non-cement handlers (NCH), consisting of office staff within the same factory. The average age of the workers were 36 years (±1.51 years), and their mean (BMI) body mass index were 21.60 (±2.56 kg/m²). The participants had been employed in the cement industry for an average of 94.15 months (±3.85 months), which is approximately 7.85 years. Cement handlers were susceptible to cement dust for about 8 hours each day, weekly 6 days. Blood specimens were collected after clotting by paramedical staff at the plant. Total of 5 ml of blood was drawn from each participant during fasting state, between 7:00 and 9:00 AM, using the vein-puncture technique. These blood samples were transferred to a plane vacutainer and quickly sent to a biochemical laboratory for analysis, where cardiovascular biomarkers were assessed. Blood glucose as well as lipid profiles in fasting state that includes total cholesterol, HDL cholesterol, triglycerides, and creatinine kinase MB, were analyzed using a fully automated turbochem biochemistry analyzer manufactured by CPC Diagnostics, USA. The troponin-I levels were measured through the ELISA method, utilizing a kit produced by Moonblind Incorporation, USA. The LDL cholesterol and VLDL cholesterol values were computed employing Fried Ewald's formula.

Exclusion criteria: Workers with a history of blood transfusions, alcohol use, cigarette and shisha smoking, anemia, asthma, cardiovascular illness, or cancer were not allowed to work. In order to lessen the impact of obesity on type 2 diabetes and pre-diabetes incidence, with a BMI exceeding 30 kg/m² were excluded from the study. Participants in this study were also excluded if they had ever worked in any other business that emits dust or fumes.

Ethical clearance: The Department of Research and Development's Ethical Committee and Review Board of Chandigarh University in Chandigarh fully authorized the protocol. Researched performed compliance with the ethical criteria that are comparable to the "1964 Declaration of Helsinki" and its later revisions.[12] [ERB/2015/17, Reference No.: DRB-PUC.] In Kalyanpur, Banjari, Bihar, the Dalmia Cement Factory's management authority obtained prior consent. Every participant was informed of the goal of the study. Every participant completed an informed consent form and willingly participated in the study. Researchers assured them of the confidentiality of their personal information, and coding was completed thereafter.

Statistical analysis: Full filled using ANOVA or the Student's paired t-test to compare two groups based on paired data at different significance levels. The data of determinations were expressed using mean ± S.E. A probability value that was statistically significant ($p < 0.05$) was taken into consideration.

Data collection: This is case-referent study; information was obtained from employees through in-person interviews that were conducted in both their native tongue and English. Employees who met the inclusion criteria were informed about the research objectives, and then they then submitted necessary data to full fill survey.[13] Workers participating in tasks such as bagging, loading, grinding, and crushing were exposed to the greatest quantities of cement dust in their immediate area.

RESULTS & DISCUSSION

The anthropometric index (BMI, Weight & Height) and mean age of the cement dust exposed cement handlers (CH) assigned as test groups and non-cement handlers (NCH) assigned as control groups were shown in Table 1.

An analysis of anthropometric measurements in cement handlers (CH) Significant differences ($P<0.05$) were seen in BMI and waist-to-hip ratio (WHR) between cement handlers and non-cement handlers. Blood samples from both cement dust-exposed and non-exposed personnel were analyzed to assess the effect of prolonged cement dust susceptibility on cardiovascular markers. Analysis measured troponin-I, creatinine kinase-MB, lipid profiles, (FBG) fasting blood glucose, and the atherogenic index of plasma. Results demonstrated substantially increased values ($P<0.05$) of FBG, total cholesterol (TC), LDL-cholesterol

(LDL-C), VLDL-cholesterol (VLDL-C), and triglycerides (TG), troponin-I, creatinine kinase-MB (CK-MB), Plasma atherogenic index and total cholesterol: HDL-cholesterol ratio in cement handlers compared to non-cement handlers (controls). The data is presented in Table 2.

Table 1:- The socio-demographic profiles of 120 individuals, consisting of Cement Handlers (CH) and Non-Cement Handlers (NCH) at Dalmia Bharat Cement Plant, were examined.

Variables	NCH (n=60) (Range)	CH (n=60) (Range)	P-Value
Age (Year)	34.07 ± 2.89 (22.0 - 42.0)	38.94 ± 2.89 (27.0 – 47.0)	> 0.05*
Weight (Kilogram)	59.57 ± 2.40 (48.0 – 69.0)	62.08 ± 5.2 (50.0 – 70.0)	< 0.05
Height (Centimeter)	159.70 ± 5.71 (142.0 – 178.0)	164.60 ± 4.97 (146.0 – 169.0)	< 0.05*
BMI (meter/Kilogram^2)	24.05 ± 1.99 (20.5 – 24.5)	22.55 ± 2.05 (18.5 – 24.5)	< 0.05*
Ratio of Waist to Hip (centimeter)	79.8 ± 3.05 (< 94.0)	90.2 ± 3.60	< 0.05*

Note: * Significant at $p < 0.05$; ** Significant at $p < 0.01$; values are given as Mean ± S.E.
Abbreviation: BMI = Body Mass Index.

Table 2:- Susceptibility to dust on fasting blood glucose (FBG), lipid profile, troponin-I, creatinine kinase MB, and the atherogenic index of plasma (AIP) was assessed among Non-Cement Handlers (NCH) and Cement Handlers (CH) with varying susceptibility durations at Dalmia Bharat Cement Plant.

Parameters	NCH (Range) (n=60)	CH (Range) (n=60)	P – Value
FBG (m mol/L)	4.73 ±0.29 (3.9 – 5.6)	5.98 ± 0.38	< 0.01**
Total cholesterol (m mol/L)	3.71 ± 0.96 (0.00 – 5.0)	5.12 ± 2.10	<0.001***
HDL-cholesterol (m mol/L)	1.50 ± 0.15 (> 1.00)	1.31 ± 0.21	< 0.05*
LDL-cholesterol (m mol/L)	1.91 ± 0.93 (0.00 – 4.0)	2.87 ± 2.04	< 0.01**
VLDL-cholesterol (m mol/L)	0.48 ± .023 (< 0.34)	0.74 ± 0.30	<0.001***
Triglycerides (m mol/L)	1.07 ± 0.50 (< 1.70)	1.64 ± 0.66	<0.001***
Troponin-I (ng/ml)	0.08 ± 0.02 (0.00 – 0.04)	0.20 ± 0.4	<0.001***
Creatinine Kinase MB (IU/L)	16.1 ± 4.48 (0.00 – 0.04)	22.2 ± 11.34	<0.001***
Atherogenic Index of Plasma	0.13 ± 0.25 (<0.11)	0.021 ± 0.24	<0.05*
Total Cholesterol: HDL- Cholesterol ratio	2.99 ± 1.17 (3.5: 1)	3.65 ± 1.75	<0.05*

Note: Data are shown as Mean ± S.E., with * indicating significance at $p < 0.05$, ** at $p < 0.01$, and *** at $p < 0.001$.
Abbreviations: FBG stands for Fasting Blood Glucose, HDL for High-Density Lipoprotein, and LDL for Low-Density Lipoprotein. VLDL = Very Low-Density Lipoprotein; Troponin-I = Trop-I.

Blood samples taken from CH exposed to cement dust and those were not analyzed to evaluate the impact of durations of exposure on fasting blood glucose (FBG), cholesterol levels (total, HDL, LDL, VLDL), triglycerides, troponin-I, creatinine kinase-MB, and atherogenic plasma's index. Additionally, the total cholesterol: HDL-cholesterol ratio (TC: HDL-C) was also evaluated. A comparative analysis based on varying periods of cement dust exposure revealed that mean values of FBG, TC, LDL-C, VLDL-C, TG, Troponin-I, CK-MB, AIP, and the TC:HDL-C ratio were significantly higher ($P < 0.05$) in cement handlers exposed to longer durations of dust exposure. The test group had the greatest mean value with an exposure period of 94.15 ± 3.85 months. The data is presented in Table 3.

Table 3:- The effect of susceptibility to dust exposure on fasting blood glucose, lipid profile, troponin-I, creatinine kinase-MB, atherogenic index of plasma, and total cholesterol: HDL-cholesterol ratio was evaluated among Cement Handlers with varying exposure durations (in months) at Dalmia Bharat Cement Plant.

Parameters	Exposure duration (months)			P - Value
	66.46 ±3.54, n=60	73.42 ± 1.42, n=60	73.42 ± 1.42, n=60	
FBG (m mol/L)	5.32 ± 0.30	5.67 ± 0.35	5.98 ± 0.38	<0.01**
Total cholesterol (m mol/L)	3.91 ± 0.85	4.57 ± 0.65	5.12 ± 0.70	<0.001***
HDL-cholesterol (m mol/L)	1.46 ± 0.1	1.49 ± 0.17	1.31 ± 0.21	< 0.05*
LDL cholesterol (m mol/L)	2.13 ± 1.38	2.44 ± 1.91	2.87 ± 2.04	< 0.05*
VLDL-cholesterol (m mol/L)	0.61 ± 0.24	0.66 ± 0.21	0.74 ± 0.30	<0.001***
Triglycerides (m mol/L)	1.21 ± 0.64	1.48 ± 0.66	1.64 ± 0.44	<0.001***
Troponin-I (ng/ml)	0.13 ± 0.19	0.16 ± 0.19	0.20 ± 0.40	<0.001***
Creatinine Kinase MB (IU/L)	18.1 ± 1.21	20.3 ± 1.29	22.2 ± 1.34	<0.001***
Atherogenic Index of Plasma	0.012 ± 0.14	0.016 ± 0.19	0.021 ± 0.24	<0.05*
Total Cholesterol: HDL- Cholesterol ratio	3.01 ± 1.11	3.23 ± 1.29	3.65 ± 1.75	<0.05*

Note: Data are shown as Mean ± S.E., with * indicating significance at $p < 0.05$, ** at $p < 0.01$, and *** at $p < 0.001$.
Abbreviations: Fasting blood glucose (FBG), high-density lipoprotein (HDL), and low-density lipoprotein (LDL), (VLDL) Very Low-Density Lipoprotein; Troponin-I (Trop-I).

Assessment of percentage of cement handlers prone to heart problem respective of CRP, CKMB, AIP and TC: HDL-C ratios were done. The data are seen as in Table 4. The atherogenic index of plasma and TC: HDL-C ratios were significantly higher in a larger percentage of cement

handlers compared to non-cement handlers. In addition to it significantly high percentage of cement handlers show considerable high values of C-reactive protein and creatinine kinase MB in their blood samples in contrast to their equivalent non-cement handlers (control). This the present study shows exposed workers are on verge of developing cardiac or cardio vascular pathogenesis.

Table 4:- Impact of cement dust in relation to higher cardiac risk factors in percentage.

Cardiac risk factors	NCH (n=60) % (n)	CH (n=60) % (n)	P-Valve
C-Reactive Protein	0 % (0)	12 % (19)	<0.01**
Creatinine Kinase MB	0 % (0)	56 % (26)	<0.01**
Atherogenic Index of Plasma	17.5 % (1)	32 % (8)	< 0.05*
Total Cholesterol: HDL-Cholesterol ratio	7.5 % (1)	14 % (7)	< 0.05*

Note: Data are shown as Mean ± S.E. With *p < 0.05; **p < 0.01; ***p < 0.001

Abbreviation: BGF = Blood Glucose Fasting; HDL = High Density Lipoprotein; LDL = Low Density Lipoprotein; VLDL = Very Low Density Lipoprotein; trop-I =Troponin-I

Cardiovascular diseases, once primarily associated with sedentary lifestyles, have now become the most prevalent metabolic condition worldwide, with rising incidence across all socioeconomic groups.[14] As best to knowledge, this solitary conducted in Bihar, state in India, that investigates pollution of air and environmental health maladies among Dalmia Bharat cement Plant workers, focusing on linking prolonged workplace exposure dust susceptibility with occurrence of cardiovascular diseases. The present study found a positive correlation, suggesting that increasing environmental pollution emerging as significant role for cardiovascular conditions. A key strength of this research is its novel approach in assessing the long-term and detrimental cardiovascular effects of cement dust exposure, an area that has not been sufficiently explored. The susceptibility aligned with heightened free radical-mediated stress, weakness, antioxidant activity, that lead to adiposis[15] and, therefore, dyslipidemia[16]. This study assesses various Vascular risk agents that's include in lipid panel, cholesterol - total HDL-Cholesterol ratio, plasma atherogenic index, fasting blood sugar, creatinine kinase-MB, troponin-I, and socio-demographic characteristics. Butt load of cholesterol- total, triglycerides, LDL/ cholesterol, and VLDL/cholesterol marked in cement handlers in contrast to non-cement handlers. Cement carries combination of heavy metals[17] and other harmful dioxins, which disrupt metabolism of fat by inhibiting lipoprotein lipase[18]. This interference results in reduced clearance of VLDL-cholesterol and chylomicrons, leading to elevated triglyceride levels. The elevated triglyceride levels observed in cement workers in this study could be attributed to their exposure to heavy metals like cadmium, lead, and mercury found in cement dust. These metals have been linked to various adverse health effects, including dyslipidemia, which occurs primarily through lipid peroxidation. This process can interfere with the enzymes that regulate cholesterol metabolism, potentially leading to increment of total and LDL plus total cholesterol levels seen in handlers in this investigation. Ours results align to previous studies[19-21] that reported elevated triglyceride, total, and LDL cholesterol levels in workers in the cement industry internationally. In comparison to another study, cement handlers in this investigation exhibited lower HDL cholesterol levels.[22] Workers with the longest exposure to cement dust showed the highest mean ratios of total to LDL cholesterol. This shows a dose-response association between exposure time and these lipid markers. Hyperlipidemia is a significant factor in the development of atherosclerosis and cardiovascular disease.[23] In our study, exposure to cement dust, which contributes to environmental pollution, is increasingly recognized as a causation factor for dyslipidemia. Increment in LDL and cholesterol-total cholesterol levels are significant risk factors for cardiovascular disease. The unfavorable changes in the lipid profile, along with the higher atherogenic index and Total-C: HDL-C ratio observed in cement workers, suggest an increased risk of cardiovascular conditions. Additionally, cement handlers in this study had higher levels of both CKMB and troponin I compared to the control group. Susceptibility to dust is linked to chronic respiratory issues, which can reduce the oxygen supply to the heart, potentially resulting in myocardial damage.[24] Cement dust contributes to changes in cardiovascular parameters by inducing oxidative stress and lowering cardiac output.[25] The presence of cement containing cadmium heavy metal exacerbates free-radical mediated stress and induces cardio toxicity, as reflected increases in CKMB and troponin I levels.[26] Although cement handlers exhibited higher levels of troponin I compared to non-cement handlers, all values remained inside the normal reference range. However, major proportion of cement handlers (66%) had creatinine-CK levels above the normal range. This implies the existence of moderate cardiac

damage and puts cement handlers at a higher risk of acquiring myocardial injury in the future. The positive relationship between the Total-C: HDL-C ratio and CKMB in cement handlers supports the notion that a higher Total-C: HDL-C ratio is a substantial risk factor for mild myocardial damage, potentially leading to myocardial infarction. Similar research found that the Total-C: HDL-C ratio and CKMB were positively related in individuals with unstable angina.[27] It was noted that cement handlers exhibit significantly higher plasma lipids and glucose levels when compared with other groups. Additionally, cardiovascular markers troponin-I and CKMB were also significantly elevated in cement handlers.

Based on the results of this study, we conclude that Cement handlers exhibited significantly dyslipidemia that include LDL cholesterol, total cholesterol, triglyceride and VLDL cholesterol,) as well as elevated cardiovascular markers (CKMB and troponin-I concentrations) compared to non-cement handlers. This suggests that cement workers may have an increased susceptibility to cardiovascular disease (CVD).

CONFLICT OF INTERESTS

There are no conflicts of interest pertaining to the publishing of this research, according to the authors.

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